GlobalLogix Retail: Enterprise Sales & Inventory Optimization

Domain: Supply Chain, Retail Operations & E-Commerce

Target Role: Data Analyst

Executive Summary:

GlobalLogix is an omnichannel consumer enterprise processing hundreds of thousands of transactions annually across diverse product lines. While top-line sales continue to expand, operating margins are under pressure due to critical supply chain bottlenecks:

1. **Stockouts on High-Velocity Goods:** Fast-moving items regularly run out of stock before warehouse teams can reorder, leading to unfulfilled demand, expedited freight costs, and lost customer goodwill.
2. **Working Capital Locked in Dead Stock:** Slow-moving and low-margin inventory occupies valuable warehouse space, increasing holding costs.
3. **Unnormalized Reporting Silos:** Transactional logs exist in wide, flat, and denormalized tables. Queries across historical volumes choke database performance, preventing real-time stock velocity monitoring and dynamic reorder alerts.

Project Objectives:

- **Data Architecture Modernization:** Transition from flat, unstructured data into a fully normalized **Third Normal Form (3NF)** relational database in MySQL.
- **Advanced Analytical Computation:** Implement production-grade SQL scripts utilizing multi-tier Common Table Expressions (CTEs) and Window Functions to compute moving demand, rolling revenues, and rank-based category performance.
- **Automated Inventory Telemetry:** Engineer dynamic operational formulas directly in SQL to track **Daily Run Rate (Sales Velocity)**, **Days of Inventory Remaining (DOIR)**, and automated **Reorder Point (ROP) Triggers**.
- **Executive Reporting:** Expose database views to Power BI to provide continuous operational visibility to warehouse managers and category leads.

Core Business Questions to Answer:

To ensure your SQL covers every line on your resume, the analysis must solve these specific business questions:

A. Revenue & Trend Dynamics (Window Functions & CTEs):

1. **Category Dominance:** What are the top 3 revenue-generating products within *each* merchandise category, and what percentage of the category's revenue do they control? (Solves via DENSE_RANK() partitioned by category)
2. **Growth Trajectory:** What is the Month-over-Month (MoM) revenue growth rate across operating quarters, and which months experienced demand contraction? (Solves via LAG() comparing current vs. previous month sales)
3. **Demand Smoothing:** What is the 7-day and 30-day moving average of order quantities per category to filter out weekly seasonality spikes? (Solves via Window Frames: ROWS BETWEEN n PRECEDING AND CURRENT ROW)

B. Inventory Velocity & Risk Detection:

4. **Sales Run Rate:** What is the actual average daily unit depletion rate for each SKU over the trailing 30 operational days?
5. **Stockout Runway (DOIR):** Based on current stock-on-hand and daily sales velocity, how many days of inventory remain before each SKU stockout occurs?
6. **Capital Exposure:** Which products are categorized as "Dead Stock" (zero sales velocity over the last 60 days but significant warehouse stock)?

C. Automated Operational Triggers (Business Logic & Alerts):

7. **Replenishment Thresholds:** Which products have breached their calculated safety stock threshold when accounting for supplier lead time:

$$\text{Reorder Alert Condition} = \text{Current Stock} \leq (\text{Daily Demand} \times \text{Lead Time Days}) + \text{Safety Stock}$$

8. **Purchase Order Recommendations:** For all items flagged as URGENT REORDER, what exact batch quantity must be purchased immediately to restore stock to a healthy 45-day operational buffer?